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Theories and applications associated with biquaternion linear canonical transform

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The quaternion linear canonical transform (QLCT) has been widely used in color image processing. Biquaternion is a more generalized algebra of quaternion, which has attracted scholars' research interest in recent years. In this paper, a new transform is proposed called the biquaternion linear canonical transforms (BiQLCTs). Due to the noncommutativity of biquaternion algebra multiplication, there are three different types of the BiQLCTs: Left-sided BiQLCT, right-sided BiQLCT, and two-side BiQLCT. The transforms are the extension of the complex linear canonical transforms. Then, the relationships between the three kinds of transforms are obtained. Next, based on the right-side biquaternion linear canonical transform (RBiQLCT), some general properties of this transform are proved. Moreover, the convolution and correlation theorems of the RBiQLCT are studied. As an application, according to the convolution operator and convolution theorem, the biquaternion linear time-invariant system is analyzed. Finally, the Heisenberg uncertainty principle for the RBiQLCT is established.

KEYWORDS

biquaternion linear canonical transforms, biquaternion linear time-invariant system, convolution theorem, quaternion linear canonical transforms, uncertainty principle

MSC CLASSIFICATION

42A38, 11R52, 94A12

1 | INTRODUCTION

Quaternions (as the Cayley–Dickson algebra of order 4) have wide applications in mathematics [1], physical [2] and signal processing [3]. Based on the quaternion algebra, some authors have generalized the linear canonical transform (LCT) to quaternion-valued function, known as the quaternionic linear canonical transform (QLCT) [4, 5]. Some useful properties of the QLCT such as linearity, reconstruction formula, continuity, boundedness, and positivity inversion formula are established in [6, 7]. Mawardi et al. [8] studied the convolution and correlation theorems and uncertainty principle of the one-sided QLCT. Li et al. [9] studied the convolution, correlation, and product theorems for the QLCT. Hu et al. [10] proposed various types of convolution formulas associated with QLCT and discussed the applications in the integral equations and the design of multiplicative filters. Moreover, Heisenberg's uncertainty principles [11, 12] for the QLCT were obtained. We [13] exploited Heisenberg's uncertainty principle associated with the quaternion windowed linear canonical transform. Some scholars have studied other types of uncertainty principles as well, such as logarithmic uncertainty principle [14], Benedicks–Amrein–Berthier uncertainty principle [15], sharp local uncertainty principle [16],

Donoho–Stark’s uncertainty principle [17], and entropic uncertainty principle [18]. As applications for the QLCT, the generalized filters and color images are introduced [19, 20]. The QLCT is becoming a very useful signal processing tool.

On the other hand, biquaternions are also known as quaternions with complex components, or complex numbers with quaternion real and imaginary parts [21]. It is another hypercomplex algebra and has a scalar, pseudoscalar, vector, and a bivector part [22]. As a generalization of the quaternions, biquaternions were first discovered by Hamilton [21] in 1853. Ward [23] introduced some properties of biquaternions. Bekar and Yayh [24] presented involutions and anti-involutions of biquaternions (complexified quaternions) and split quaternions. The fundamental properties of biquaternions were presented including several different representations in this paper [25]. Sangwine [26] derived the roots of -1 in the set of biquaternions. A biquaternion-valued signal [22], which is a quaternion with complex components signal, includes a scalar, pseudoscalar, vector, and a bivector part. The aim of our study of biquaternion signals is the geometric properties of biquaternion. These properties allow biquaternion-valued signals to process signals with several different geometric properties that may occur simultaneously [22]. Based on the concept of biquaternion, Said et al. [22] studied the extension of the Fourier transform to biquaternion-valued function and proposed the biquaternion Fourier transform (BiQFT). In this paper [22], some general properties of BiQFT were exploited, and the Hermitian symmetries of the BiQFT and their relation to the geometric nature of a biquaternion-valued function were investigated. Recently, Bi et al. [27] proposed the biquaternion Z transformation, and using the transform to solve a class of biquaternion recurrence relations.

As an extension of the quaternion Fourier transform (QFT) [28, 29], the BiQFT has been attracted more and more attention. To the best of our knowledge, researching the LCT to biquaternion-valued function has never been proposed up to date. In this paper, we propose the biquaternion linear canonical transforms (BiQLCTs) by substituting the biquaternion Fourier transform kernel with the BiQLCT kernel. According to the properties of biquaternion, the BiQLCTs have three kinds forms: Two-side biquaternion linear canonical transform (TBiQLCT), left-side biquaternion linear canonical transform (LBiQLCT), and right-side biquaternion linear canonical transform (RBiQLCT). Applying these definitions, the relations between the three kinds of transforms are obtained. Then, based on the RBiQLCT, the inversion formula and Parseval theorem of the RBiQLCT are analyzed. In addition, the biquaternion convolution and correlation operators are defined, and the convolution and correlation theorems of the RBiQLCT are studied. According to the convolution operator and convolution theorem, the biquaternion linear time-invariant system is analyzed. Finally, the Heisenberg uncertainty principle for the RBiQLCT is established.

The paper is organized as follows: Section 2 gives a brief introduction to some general definitions and basic properties of biquaternions. We study the definition and properties of the BiQLCTs in Section 3. Section 4 provides the convolution and correlation theorems associated with the RBiQLCT. The biquaternion linear time-invariant system and biquaternion filter are analyzed in Section 5. In Section 6, we obtain the Heisenberg uncertainty principle for the RBiQLCT. Some conclusions are drawn in Section 7,

2 | PRELIMINARIES

In this section, we mainly review some basic facts regarding biquaternions. Biquaternions, also known as complexified quaternions, may be represented as quaternions with complex coefficients [21]. It forms an eight-dimensional algebra first discovered by Hamilton in 1853 [21].

The set of biquaternions is generally denoted $\mathbb{H}_{\mathbb{C}}$ [23, 24]. A biquaternion $q \in \mathbb{H}_{\mathbb{C}}$ can be written in the form

$$q = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}, \quad (1)$$

where $q_0, q_1, q_2, q_3 \in \mathbb{C}$ are complex numbers and $\mathbf{i}, \mathbf{j}, \mathbf{k}$ are exactly the same in real quaternions. If $q_0 = 0$, then biquaternion q is known as pure biquaternion [25].

The complex numbers are written by the notation [25]

$$\mathbf{I}^2 = -1. \quad (2)$$

q is also possible to write a biquaternion in the following form [26]:

$$q = S(q) + V(q), \quad (3)$$

where $S(q) = q_0$ is the (complex-valued) scalar part of q and $V(q) = q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}$ is its (complex quaternion-valued) vector part.

Given two biquaternions $q = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k} = S(q) + V(q)$ and $p = p_0 + p_1\mathbf{i} + p_2\mathbf{j} + p_3\mathbf{k} = S(p) + V(p)$, then

$$q + p = (S(q) + S(p)) + (V(q) + V(p)), \quad (4)$$

$$\lambda q = \lambda S(q) + \lambda V(q), \quad (5)$$

$$qp = S(q)S(p) - \langle V(q), V(p) \rangle + S(q)V(p) + S(p)V(q) + V(q) \wedge V(p), \quad (6)$$

where $\lambda \in \mathbb{R}$,

$$\langle V(q), V(p) \rangle = q_1p_1 + q_2p_2 + q_3p_3,$$

$$V(q) \wedge V(p) = (q_2p_3 - q_3p_2)\mathbf{i} - (q_1p_3 - q_3p_1)\mathbf{j} + (q_1p_2 - q_2p_1)\mathbf{k}.$$

If q is orthogonal to p , then $\langle q, p \rangle = 0$.

Moreover, the real and imaginary parts (with respect to $\mathbf{I}$) of a biquaternion are respectively defined as [22, 26]

$$\Re(q) = \Re(q_0) + \Re(q_1)\mathbf{i} + \Re(q_2)\mathbf{j} + \Re(q_3)\mathbf{k}, \quad (7)$$

$$\Im(q) = \Im(q_0) + \Im(q_1)\mathbf{i} + \Im(q_2)\mathbf{j} + \Im(q_3)\mathbf{k}, \quad (8)$$

where $\Re(q)$ and $\Im(q)$ are (real) quaternions. $\Re(q_i)$ is the real part and $\Im(q_i)$ ($i = 0, 1, 2, 3$) is the imaginary part of a complex number. So any biquaternion $q \in \mathbb{H}_{\mathbb{C}}$ can be written as $q = \Re(q) + \mathbf{I}\Im(q)$. The complex imaginary unit $\mathbf{I}$ commutes with the quaternion imaginary units $\mathbf{i}, \mathbf{j}, \mathbf{k}$, that is to say [22]

$$\mathbf{iI} = \mathbf{Ii}, \mathbf{jI} = \mathbf{Ij}, \mathbf{kI} = \mathbf{Ik}. \quad (9)$$

The quaternion conjugate of a biquaternion $q \in \mathbb{H}_{\mathbb{C}}$ is [26]

$$\tilde{q} = S(q) - V(q), \quad (10)$$

The complex conjugate of a biquaternion $q \in \mathbb{H}_{\mathbb{C}}$ is defined as [22]

$$\check{q} = \check{q}_0 + \check{q}_1\mathbf{i} + \check{q}_2\mathbf{j} + \check{q}_3\mathbf{k}, \quad (11)$$

where $\check{q}_0, \check{q}_1, \check{q}_2, \check{q}_3$ are the complex conjugates of the complex coefficients of q .

Biquaternion conjugation is the combination of the two conjugations that have just been defined. The biquaternion conjugate of q is defined as [26]

$$\bar{q} = \tilde{\check{q}} = \check{q} - \check{q}_1\mathbf{i} - \check{q}_2\mathbf{j} - \check{q}_3\mathbf{k}. \quad (12)$$

Complex conjugation is multiplicative, that is, $\check{pq} = \check{p}\check{q}$, while quaternion conjugation and biquaternion conjugation are involutive, that is, $\overline{\bar{q}} = q$.

Proposition 1 ([25]). *A biquaternion does not, in general, commute with its complex conjugate: Reversing the order of the product yields a complex conjugate result.*

The norm of a biquaternion q is defined by [25]

$$\|q\|^2 = |q_0|^2 + |q_1|^2 + |q_2|^2 + |q_3|^2.$$

If $\|q\| = 1$, then q is called unit biquaternion.

An important issue in defining a biquaternion Fourier transform is to define a biquaternion-valued exponential for the kernel of the transform [22]. In the following, we will mainly consider exponential kernels containing a biquaternion root of -1 [26]. The exponential kernel is essential to many properties of the BiQFT [22].

A biquaternion $\mu \in \mathbb{H}_{\mathbb{C}}$ is a biquaternion root of -1 iff $\mu^2 = -1$ [26].

Given any biquaternion root of -1 , μ and any biquaternion $q = q_0 + q_1\mathbf{i} + q_2\mathbf{j} + q_3\mathbf{k}$, q can be rewritten as [22]

$$q = (q'_0 + q'_1\mu) + (q'_2 + q'_3\mu)\nu = q'_0 + q'_1\mu + q'_2\nu + q'_3\mu\nu, \quad (13)$$

where ν is a biquaternion root of -1 orthogonal to μ and q'_0, q'_1, q'_2, q'_3 are complex numbers. Equation (13) allows the definition of a decomposition for any biquaternion q , with respect to μ and ν , and $\text{Simp}(q) = (q'_0 + q'_1\mu)$ is simplex part, $\text{Perp}(q) = (q'_2 + q'_3\mu)\nu$ is perplex part. So $q = \text{Simp}(q) + \text{Perp}(q)$.

The exponential of a biquaternion q is defined by [22]

$$e^q = \sum_{n \in \mathbb{N}} \frac{q^n}{n!}.$$

A biquaternion-valued function $f(x)$ is given by [22]

$$\begin{aligned} f(x) &= f_0(x) + f_1\mathbf{i} + f_2\mathbf{j} + f_3\mathbf{k} \\ &= \text{Simp}(f) + \text{Perp}(f) \\ &= S(f) + V(f) \\ &= \Re(f) + \mathbf{I}\Im(f), \end{aligned} \quad (14)$$

where f_0, f_1, f_2, f_3 are complex-valued functions. These signals can represent a variety of physical quantities (such as dipoles and magnetic rings for recording electromagnetic signals) captured on sensors at the same location [22].

An inner product of biquaternion functions f, g can be defined on $L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$ by

$$(f, g)_{L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})} = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \bar{g}(x, y) dx dy.$$

3 | BIQUATERNION LINEAR CANONICAL TRANSFORM

According to the biquaternion algebra, we propose the BiQLCT in this section. Unless otherwise stated, we assume that the following function is continuous and that both the function and its BiQLCT are integrable (in the Lebesgue sense) in this paper. Moreover, it is necessary to note that the multiplication in the next integral is done from left to right because the biquaternion algebra is noncommutativity. So there are three types of the BiQLCT:

3.1 | Definitions

Definition 1 (TBiQLCT). Let $A_i = \begin{bmatrix} a_i & b_i \\ c_i & d_i \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be a matrix parameter satisfying $\det(A_i) = 1$, for $i = 1, 2$. The TBiQLCT of function $f \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$ is defined by

$$\mathcal{L}_{A_1, A_2}^{TB} \{f\}(\omega, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} K_{A_1}^{\mu}(x, \omega) f(x, y) K_{A_2}^{\nu}(y, v) dx dy. \quad (15)$$

Definition 2 (LBiQLCT). Let $A_i = \begin{bmatrix} a_i & b_i \\ c_i & d_i \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be a matrix parameter satisfying $\det(A_i) = 1$, for $i = 1, 2$. The LBiQLCT of function $f \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$ is defined by

$$\mathcal{L}_{A_1, A_2}^{LB} \{f\}(\omega, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} K_{A_1}^{\mu}(x, \omega) K_{A_2}^{\mu}(y, v) f(x, y) dx dy. \quad (16)$$

Definition 3 (RBiQLCT). Let $A_i = \begin{bmatrix} a_i & b_i \\ c_i & d_i \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be a matrix parameter satisfying $\det(A_i) = 1$, for $i = 1, 2$. The RBiQLCT of function $f \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$ is defined by

$$\mathcal{L}_{A_1, A_2}^{RB}\{f\}(\omega, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) K_{A_1}^{\mu}(x, \omega) K_{A_2}^{\nu}(y, v) dx dy, \quad (17)$$

where μ and ν in the TBiQLCT are two pure unit biquaternions that are orthogonal to each other, $(x, y), (\omega, v) \in \mathbb{R}^2$ and the kernel functions

$$K_{A_i}^{\mu}(x, \omega) := \begin{cases} \frac{1}{\sqrt{2\pi|b_i|}} e^{\mu \left(\frac{a_i}{2b_i} x^2 - \frac{x\omega}{b_i} + \frac{d_i}{2b_i} \omega^2 - \frac{\pi}{4} \right)}, & b_i \neq 0 \\ \sqrt{d_i} e^{\mu \frac{c_i d_i}{2} \omega^2} \delta(x - d_i \omega), & b_i = 0 \end{cases}, \quad (18)$$

and

$$K_{A_2}^{\nu}(y, v) := \begin{cases} \frac{1}{\sqrt{2\pi|b_2|}} e^{\nu \left(\frac{a_2}{2b_2} y^2 - \frac{yv}{b_2} + \frac{d_2}{2b_2} v^2 - \frac{\pi}{4} \right)}, & b_2 \neq 0 \\ \sqrt{d_2} e^{\nu \frac{c_2 d_2}{2} v^2} \delta(y - d_2 v), & b_2 = 0 \end{cases}, \quad (19)$$

$\delta(\cdot)$ representing the Dirac function.

According to the definitions of the BiQLCTs, we can obtain the relations between the BiQLCTs and the BiQFTs. For example, the relation between the RBiQLCT and the RBiQFT:

$$\mathcal{L}_{A_1, A_2}^{RB}\{f\}(\omega, v) = \frac{1}{2\pi\sqrt{|b_1 b_2|}} F_B^R \left(f(x, y) e^{\mu \left(\frac{a_1}{2b_1} x^2 + \frac{a_2}{2b_2} y^2 \right)} \right) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) e^{\mu \left(\frac{d_1}{2b_1} \omega^2 + \frac{d_2}{2b_2} v^2 - \frac{\pi}{2} \right)}, \quad (20)$$

where the definition of the RBiQFT is the following formula:

$$F_B^R(f)(\omega, v) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) e^{-\mu(\omega x + v y)} dx dy. \quad (21)$$

Note that to every different chosen there corresponds a different transform.

- If μ, ν are **I**, then the BiQLCTs become to the complex LCTs for the components of the biquaternion function [22].
- If μ, ν are quaternions root of -1 , then the BiQLCTs become to the quaternion LCTs for the real and imaginary parts of the biquaternion function [22].

According to the three definitions of the BiQLCT, we can obtain the relationships between the TBiQLCT, LBiQLCT, and RBiQLCT. Through the relationships between the three transforms, we can effectively implement another transform based on the algorithm of one transform.

3.2 | Relationship between LBiQLCT and RBiQLCT

Theorem 2. The LBiQLCT and RBiQLCT are related by the following equation:

$$\mathcal{L}_{A_1, A_2}^{RB}\{f\}(\omega, v) = \mathcal{L}_{A_1, A_2}^{LB}\{Simp(f)\}(\omega, v) - \mathcal{L}_{A_1', A_2'}^{LB}\{Perp(f)\}(\omega, v), \quad (22)$$

where $A_i' = \begin{bmatrix} a_i & -b_i \\ -c_i & d_i \end{bmatrix} \in \mathbb{R}^{2 \times 2}, (i=1, 2)$.

Proof. According to Equation (14), we obtain

$$\begin{aligned} f(x, y)K_{A_1}^{\mu}(x, \omega)K_{A_2}^{\mu}(y, v) &= (\text{Simp}(f) + \text{Perp}(f))K_{A_1}^{\mu}(x, \omega)K_{A_2}^{\mu}(y, v) \\ &= K_{A_1}^{\mu}(x, \omega)K_{A_2}^{\mu}(y, v)\text{Simp}(f) - K_{A_1'}^{\mu}(x, \omega)K_{A_2'}^{\mu}(y, v)\text{Perp}(f). \end{aligned} \quad (23)$$

Integrating both sides with respect to $dx dy$, we have

$$\mathcal{L}_{A_1, A_2}^{RB}\{f\}(\omega, v) = \mathcal{L}_{A_1, A_2}^{LB}\{\text{Simp}(f)\}(\omega, v) - \mathcal{L}_{A_1', A_2'}^{LB}\{\text{Perp}(f)\}(\omega, v). \quad (24)$$

□

3.3 | Relationship between TBiQLCT and LBiQLCT

Theorem 3. *The TBiQLCT and LBiQLCT are related by the following equation:*

$$\mathcal{L}_{A_1, A_2}^{TB}\{f\}(\omega, v) = \mathcal{L}_{A_1, A_2}^{LB}\{f\}(\omega, v) + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (f v - \mu f) \cos \Phi_1 \sin \Phi_2 + (f + \mu f v) \sin \Phi_1 \sin \Phi_2 dx dy. \quad (25)$$

Proof. Let $\Phi_i = \frac{a_i}{2b_i}x^2 - \frac{x\omega}{b_i} + \frac{d_i}{2b_i}\omega^2 - \frac{\pi}{4}$, then $e^{i\mu\Phi_i} = \cos \Phi_i + \mu \sin \Phi_i$.

So, we obtain

$$\begin{aligned} K_{A_1}^{\mu}(x, \omega)f(x, y)K_{A_2}^{\mu}(y, v) &= \frac{1}{2\pi\sqrt{|b_1 b_2|}}(\cos \Phi_1 + \mu \sin \Phi_1)f(x, y)(\cos \Phi_2 + v \sin \Phi_2) \\ &= K_{A_1}^{\mu}(x, \omega)K_{A_2}^{\mu}(y, v)f(x, y) - \mu f \cos \Phi_1 \sin \Phi_2 \\ &\quad + f \sin \Phi_1 \sin \Phi_2 + f v \cos \Phi_1 \sin \Phi_2 + \mu f v \sin \Phi_1 \cos \Phi_2 \\ &= K_{A_1}^{\mu}(x, \omega)K_{A_2}^{\mu}(y, v)f(x, y) + (f v - \mu f) \cos \Phi_1 \sin \Phi_2 + (f + \mu f v) \sin \Phi_1 \cos \Phi_2. \end{aligned} \quad (26)$$

According to (6) and integrating both sides with respect to $dx dy$, we have

$$\mathcal{L}_{A_1, A_2}^{TB}\{f\}(\omega, v) = \mathcal{L}_{A_1, A_2}^{LB}\{f\}(\omega, v) + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (f v - \mu f) \cos \Phi_1 \sin \Phi_2 + (f + \mu f v) \sin \Phi_1 \sin \Phi_2 dx dy. \quad (27)$$

□

3.4 | Relationship between TBiQLCT and RBiQLCT

Applying the method of Theorem 3, the relationship between TBiQLCT and RBiQLCT can be obtained.

Theorem 4. *The TBiQLCT and RBiQLCT are related by the following equation:*

$$\begin{aligned} \mathcal{L}_{A_1, A_2}^{TB}\{f\}(\omega, v) &= \mathcal{L}_{A_1, A_2}^{RB}\{f\}(\omega, v) + \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y)(v - \mu) \cos \Phi_1 \sin \Phi_2 \\ &\quad + (f(x, y) + \mu f(x, y)v) \sin \Phi_1 \sin \Phi_2 + 2\mu V(f(x, y)) \sin \Phi_1 \cos \Phi_2 dx dy. \end{aligned} \quad (28)$$

The relationships between these three transforms are very important in fast algorithms for calculating the BiQLCTs. For example, if we want to compute the LBiQLCT and RBiQLCT, we can first compute the LBiQLCT and then use Theorem 2 to obtain the implementation of the RBiQLCT.

Next, we limit the transform considered to the case of the RBiQLCT, and other transforms have similar properties.

Theorem 5 (Inversion formula of the RBiQLCT). *Let $f \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$. Then we have the inversion formula of the RBiQLCT,*

$$\begin{aligned}
 f(x, y) &= \left(\mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) \right)^{-1} \left[\mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) \right] (x, y) \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) K_{A_1^{-1}}^{\mu}(\omega, x) K_{A_2^{-1}}^{\mu}(v, y) d\omega dv,
 \end{aligned} \tag{29}$$

where $A_i^{-1} = \begin{bmatrix} d_i & -b_i \\ -c_i & a_i \end{bmatrix} \in \mathbb{R}^{2 \times 2}$, for $i = 1, 2$.

Proof. Based on the definition of the RBiQLCT, we have

$$\begin{aligned}
 & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) K_{A_1^{-1}}^{\mu}(\omega, x) K_{A_2^{-1}}^{\mu}(v, y) d\omega dv \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x', y') \frac{1}{\sqrt{2\pi|b_1|}} e^{\mu\left(\frac{a_1}{2b_1}x'^2 - \frac{x'\omega}{b_1} + \frac{d_1}{2b_1}\omega^2 - \frac{\pi}{4}\right)} \\
 & \quad \times \frac{1}{\sqrt{2\pi|b_2|}} e^{\mu\left(\frac{a_2}{2b_2}y'^2 - \frac{y'v}{b_2} + \frac{d_2}{2b_2}v^2 - \frac{\pi}{4}\right)} dx' dy' \frac{1}{\sqrt{2\pi|b_1|}} e^{\mu\left(\frac{a_1}{-2b_1}x^2 - \frac{x\omega}{-b_1} + \frac{d_1}{-2b_1}\omega^2 - \frac{\pi}{4}\right)} \\
 & \quad \times \frac{1}{\sqrt{2\pi|b_2|}} e^{\mu\left(\frac{a_2}{-2b_2}y^2 - \frac{yv}{-b_2} + \frac{d_2}{-2b_2}v^2 - \frac{\pi}{4}\right)} d\omega dv \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x', y') e^{\mu\left(\frac{a_1}{2b_1}(x-x'^2)\right)} \delta(x-x'^2) e^{\mu\left(\frac{a_2}{2b_2}(y-y'^2)\right)} \delta(y-y'^2) dx' dy' \\
 &= f(x, y).
 \end{aligned} \tag{30}$$

□

Inverse transform is a very important tool in signal processing, which is used in convolution theorem, correlation theorem, uncertainty principle, Plancherel theorem, sampling, and so on. The inverse transforms of the BiQLCTs can be thought of as more general forms of the QLCTs. The inverse transforms of the BiQLCTs will provide theoretical basis for future research.

Let us prove Plancherel formula of RBiQLCT as follows:

Theorem 6 (Plancherel theorem of the RBiQLCT). *Let $f, g \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$. Then we have*

$$\left(\mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v), \mathcal{L}_{A_1, A_2}^{RB} \{g\}(\omega, v) \right) = (f, g). \tag{31}$$

Proof. Applying Theorem 5 and the definition of the RBiQLCT, we obtain

$$\begin{aligned}
 \left(\mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v), \mathcal{L}_{A_1, A_2}^{RB} \{g\}(\omega, v) \right) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) \widetilde{\mathcal{L}_{A_1, A_2}^{RB} \{g\}(\omega, v)} d\omega dv \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) K_{A_2^{-1}}^{\mu}(v, y) K_{A_1^{-1}}^{\mu}(\omega, x) d\omega dv \tilde{g}(x, y) dx dy \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x, y) \tilde{g}(x, y) dx dy \\
 &= (f, g).
 \end{aligned} \tag{32}$$

□

Corollary 7. If $f = g$, then

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left| \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) \right|^2 d\omega dv = \|f\|_{L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})}^2. \quad (33)$$

4 | CONVOLUTION THEOREM

In this section, the convolution theorem of the RBiQLCT is exploited.

Definition 4. Let $f, g \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$, we define the convolution operator of the RBiQLCT as follows:

$$(f \ast g)(x, y) = \left[f(x, y) \ast \left(g(x, y) e^{\mu\left(\frac{a_1}{2b_1}x^2 + \frac{a_2}{2b_2}y^2\right)} \right) \right] e^{-\mu\left(\frac{a_1}{2b_1}x^2 + \frac{a_2}{2b_2}y^2\right)}, \quad (34)$$

where $\ast$ is the traditional convolution operator.

According to the convolution operator of the RBiQLCT, we have the following convolution theorem of the RBiQLCT.

Theorem 8. Assume $f, g \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$, then the RBiQLCT of the convolution of f and g are provided by

$$\mathcal{L}_{A_1, A_2}^{RB} \{f \ast g\}(\omega, v) = F_B^R(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \mathcal{L}_{A_1, A_2}^{RB} \{Simp(g)\}(\omega, v) + F_B^R(f) \left(-\frac{\omega}{b_1}, -\frac{v}{b_2} \right) \mathcal{L}_{A_1, A_2}^{RB} \{Perp(g)\}(\omega, v). \quad (35)$$

Proof. According to the definition of the RBiQLCT, we obtain

$$\begin{aligned} \mathcal{L}_{A_1, A_2}^{RB} \{f \ast g\}(\omega, v) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} (f \ast g)(x, y) K_{A_1}^{\mu}(x, \omega) K_{A_2}^{\mu}(y, v) dx dy \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\tau, \eta) g(x - \tau, y - \eta) e^{\mu\left(\frac{a_1}{2b_1}(x-\tau)^2 + \frac{a_2}{2b_2}(y-\eta)^2\right)} d\tau d\eta \\ &\quad \times e^{-\mu\left(\frac{a_1}{2b_1}x^2 + \frac{a_2}{2b_2}y^2\right)} \frac{1}{\sqrt{2\pi|b_1|}} e^{\mu\left(\frac{a_1}{2b_1}x^2 - \frac{x\omega}{b_1} + \frac{d_1}{2b_1}\omega^2 - \frac{\pi}{4}\right)} \frac{1}{\sqrt{2\pi|b_2|}} e^{\mu\left(\frac{a_2}{2b_2}y^2 - \frac{yv}{b_2} + \frac{d_2}{2b_2}v^2 - \frac{\pi}{4}\right)} dx dy. \end{aligned} \quad (36)$$

Let $\epsilon = x - \tau$, $\rho = y - \eta$, then the above formula becomes

$$\begin{aligned} \mathcal{L}_{A_1, A_2}^{RB} \{f \ast g\}(\omega, v) &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\tau, \eta) g(\epsilon, \rho) e^{\mu\left(\frac{a_1}{2b_1}\epsilon^2 + \frac{a_2}{2b_2}\rho^2\right)} d\tau d\eta \frac{1}{\sqrt{2\pi|b_1|}} e^{\mu\left(-\frac{\omega}{b_1}(\epsilon+\tau) + \frac{d_1}{2b_1}\omega^2 - \frac{\pi}{4}\right)} \frac{1}{\sqrt{2\pi|b_2|}} e^{\mu\left(-\frac{v}{b_2}(\rho+\eta) + \frac{d_2}{2b_2}v^2 - \frac{\pi}{4}\right)} d\epsilon d\rho \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\tau, \eta) (Simp(g) + Perp(g))(\epsilon, \rho) e^{-\mu\left(\frac{\omega\tau}{b_1} + \frac{v\eta}{b_2}\right)} K_{A_1}^{\mu}(\epsilon, \omega) K_{A_2}^{\mu}(\rho, v) d\epsilon d\rho d\tau d\eta \\ &= F_B^R(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \mathcal{L}_{A_1, A_2}^{RB} \{Simp(g)\}(\omega, v) + F_B^R(f) \left(-\frac{\omega}{b_1}, -\frac{v}{b_2} \right) \mathcal{L}_{A_1, A_2}^{RB} \{Perp(g)\}(\omega, v). \end{aligned} \quad (37)$$

In the case where $f(x, y)$ is the even function □

$$f(x, y) = f(-x, -y), \quad (38)$$

then according to the definition of the RBiQFT, the RBiQFT of $f(x, y)$ is also the even function

$$F_B^R(f)(\omega, v) = F_B^R(f)(-\omega, -v), \quad (39)$$

and the formula (35) can be simplified

$$\mathcal{L}_{A_1, A_2}^{RB}\{f * g\}(\omega, v) = F_B^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) \mathcal{L}_{A_1, A_2}^{RB}\{g\}(\omega, v). \quad (40)$$

Hence, when $f(x, y)$ is even, the convolution operation of two biquaternion-valued functions in the time domain is equivalent to the product operation in the frequency domain.

When $f(x, y)$ is the odd function

$$f(x, y) = -f(-x, -y), \quad (41)$$

then

$$F_B^R(f)(\omega, v) = -F_B^R(f)(-\omega, -v), \quad (42)$$

and the formula (35) can be simplified

$$\mathcal{L}_{A_1, A_2}^{RB}\{f * g\}(\omega, v) = F_B^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) \mathcal{L}_{A_1, A_2}^{RB}\{g'\}(\omega, v). \quad (43)$$

where $g'(x, y) = \text{Simp}(g) - \text{Perp}(g)$.

In general, if $f(x, y)$ is neither even nor odd, based on (40) and (43), we can conclude that the relation between the inputs and the output of the convolution for the RBiQLCT in the frequency domain can be written as

$$\mathcal{L}_{A_1, A_2}^{RB}\{f * g\}(\omega, v) = F_{e,B}^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) \mathcal{L}_{A_1, A_2}^{RB}\{g\}(\omega, v) + F_{o,B}^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) \mathcal{L}_{A_1, A_2}^{RB}\{g'\}(\omega, v), \quad (44)$$

where $F_{e,B}^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right)$ and $F_{o,B}^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right)$ are the even and odd parts of the RBiQLCT $F_B^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right)$

$$F_{e,B}^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) = \frac{\left[F_B^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) + F_B^R(f)\left(-\frac{\omega}{b_1}, -\frac{v}{b_2}\right)\right]}{2}, \quad (45)$$

$$F_{o,B}^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) = \frac{\left[F_B^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) - F_B^R(f)\left(-\frac{\omega}{b_1}, -\frac{v}{b_2}\right)\right]}{2}. \quad (46)$$

If $A_1 = A_2 = (0, 1, -1, 0)$, then the RBiQLCT will be reduced to the right-side BiQFT (RBiQFT), and we can obtain the following relationship:

$$\mathcal{L}_{A_1, A_2}^{RB}\{f\}(\omega, v) = \frac{1}{2\pi} F_B^R(f)(\omega, v) \cdot \frac{1}{\mu}. \quad (47)$$

Then the convolution operator (34) reduced to the traditional convolution operator

$$(f * g)(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\tau, \eta) g(x - \tau, y - \eta) d\tau d\eta, \quad (48)$$

based on the above convolution operator (48), we can obtain the convolution theorem of the RBiFT:

$$F_B^R\{f * g\}(\omega, v) = F_B^R(f)(\omega, v) F_B^R\{\text{Simp}(g)\}(\omega, v) + F_B^R(f)(-\omega, -v) F_B^R\{\text{Perp}(g)\}(\omega, v). \quad (49)$$

When $f(x, y) = f(-x, -y)$, then the convolution theorem of the RBiQFT can be simplified

$$F_B^R\{f * g\}(\omega, v) = F_B^R\{f * g\}(\omega, v) = F_{e,B}^R(f)(\omega, v) F_B^R\{g\}(\omega, v). \quad (50)$$

When $f(x, y) = -f(-x, -y)$, then the formula (49) can be simplified

$$F_B^R\{f * g\}(\omega, v) = F_B^R\{f * g\}(\omega, v) = F_{o,B}^R(f)(\omega, v) F_B^R\{g'\}(\omega, v). \quad (51)$$

In general, if $f(x, y)$ is neither even nor odd, based on (50) and (51), we can conclude that the relation between the inputs and the output of the convolution for the RBiQFT in the frequency domain can be written as

$$F_B^R\{f * g\}(\omega, v) = F_{e,B}^R(f)(\omega, v)F_B^R\{g\}(\omega, v) + F_{o,B}^R(f)(\omega, v)F_B^R\{g'\}(\omega, v). \quad (52)$$

According to (48), if $f(x, y)$ and $g(x, y)$ are even or odd, then $(f * g)(x, y)$ is even, and if $f(x, y)$ is even and $g(x, y)$ is odd (or $f(x, y)$ is odd and $g(x, y)$ is even), then $(f * g)(x, y)$ is odd. So we obtain

$$F_{e,B}^R\{f * g\}(\omega, v) = F_{e,B}^R(f)(\omega, v)F_{e,B}^R\{g\}(\omega, v) + F_{o,B}^R(f)(\omega, v)F_{o,B}^R\{g'\}(\omega, v), \quad (53)$$

$$F_{o,B}^R\{f * g\}(\omega, v) = F_{e,B}^R(f)(\omega, v)F_{o,B}^R\{g\}(\omega, v) + F_{o,B}^R(f)(\omega, v)F_{e,B}^R\{g'\}(\omega, v). \quad (54)$$

5 | CORRELATION THEOREM

In this section, the correlation theorem of the RBiQLCT is derived.

Definition 5. Let $f, g \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$, we define the correlation operator of the RBiQLCT as follows:

$$(f \star g)(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x + \tau, y + \eta) \overline{g(\tau, \eta)} e^{-\mu\left(\frac{a_1}{2b_1}\tau^2 + \frac{a_2}{2b_2}\eta^2\right)} d\tau d\eta \cdot e^{-\mu\left(\frac{d_1}{2b_1}x^2 + \frac{d_2}{2b_2}y^2\right)}. \quad (55)$$

According to the correlation operator of the RBiQLCT, we have the following correlation theorem of the RBiQLCT.

Theorem 9. Assume $f, g \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$, then the RBiQLCT of the correlation of f and g are provided by

$$\begin{aligned} & \mathcal{L}_{A_1, A_2}^{RB}\{f \star g\}(\omega, v) \\ &= \left\{ F_B^R(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \mathcal{L}_{A'_1, A'_2}^{RB} \left\{ \overline{\text{Simp}(g)} \right\} (-\omega, -v) + F_B^R(f) \left(-\frac{\omega}{b_1}, -\frac{v}{b_2} \right) \mathcal{L}_{A'_1, A'_2}^{RB} \left\{ \overline{\text{Perp}(g)} \right\} (-\omega, -v) \right\} e^{\mu\left(\frac{d_1}{2b_1}\omega^2 + \frac{d_2}{2b_2}v^2\right)}. \end{aligned} \quad (56)$$

Proof.

$$\begin{aligned} & \mathcal{L}_{A_1, A_2}^{RB}\{f \star g\}(\omega, v) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x + \tau, y + \eta) \overline{g(\tau, \eta)} e^{-\mu\left(\frac{a_1}{2b_1}\tau^2 + \frac{a_2}{2b_2}\eta^2\right)} d\tau d\eta \frac{1}{\sqrt{2\pi|b_1|}} e^{\mu\left(-\frac{x\omega}{b_1} + \frac{d_1}{2b_1}\omega^2 - \frac{\pi}{4}\right)} \frac{1}{\sqrt{2\pi|b_2|}} e^{\mu\left(-\frac{yv}{b_2} + \frac{d_2}{2b_2}v^2 - \frac{\pi}{4}\right)} dx dy. \end{aligned} \quad (57)$$

Let $\varepsilon = x + \tau, \zeta = y + \eta$, then the above formula becomes

$$\begin{aligned} & \mathcal{L}_{A_1, A_2}^{RB}\{f \star g\}(\omega, v) \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\varepsilon, \zeta) \overline{g(\tau, \eta)} e^{-\mu\left(\frac{a_1}{2b_1}\tau^2 + \frac{a_2}{2b_2}\eta^2\right)} d\tau d\eta \frac{1}{\sqrt{2\pi|b_1|}} e^{\mu\left(-\frac{(\varepsilon-\tau)\omega}{b_1} + \frac{d_1}{2b_1}\omega^2 - \frac{\pi}{4}\right)} \frac{1}{\sqrt{2\pi|b_2|}} e^{\mu\left(-\frac{(\zeta-\eta)v}{b_2} + \frac{d_2}{2b_2}v^2 - \frac{\pi}{4}\right)} d\varepsilon d\zeta \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\varepsilon, \zeta) (\overline{\text{Simp}(g)} + \overline{\text{Perp}(g)})(\tau, \eta) K_{A'_1}^{\mu}(\tau, -\omega) K_{A'_2}^{\mu}(\eta, -v) d\tau d\eta e^{-\mu\left(\frac{\omega\varepsilon}{b_1} + \frac{v\zeta}{b_2}\right)} e^{\mu\left(\frac{d_1}{2b_1}\omega^2 + \frac{d_2}{2b_2}v^2\right)} d\varepsilon d\zeta \\ &= \left\{ F_B^R(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \mathcal{L}_{A'_1, A'_2}^{RB} \left\{ \overline{\text{Simp}(g)} \right\} (-\omega, -v) + F_B^R(f) \left(-\frac{\omega}{b_1}, -\frac{v}{b_2} \right) \mathcal{L}_{A'_1, A'_2}^{RB} \left\{ \overline{\text{Perp}(g)} \right\} (-\omega, -v) \right\} e^{\mu\left(\frac{d_1}{2b_1}\omega^2 + \frac{d_2}{2b_2}v^2\right)}. \end{aligned} \quad (58)$$

□

In general, if $f(x, y)$ is neither even nor odd, we obtain

$$\begin{aligned} & \mathcal{L}_{A_1, A_2}^{RB} \{f \star g\}(\omega, v) \\ &= \left[F_{e, B}^R(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \mathcal{L}_{A'_1, A'_2}^{RB} \{\bar{g}\}(\omega, v) + F_{o, B}^R(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \mathcal{L}_{A'_1, A'_2}^{RB} \{ \overline{\text{Simp}(g)} - \overline{\text{Perp}(g)} \}(\omega, v) \right] e^{\mu \left(\frac{d_1}{2b_1} \omega^2 + \frac{d_2}{2b_2} v^2 \right)}. \end{aligned} \quad (59)$$

If $A_1 = A_2 = (0, 1, -1, 0)$, then the correlation operator reduced to the traditional correlation operator

$$(f \tilde{\star} g)(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x + \tau, y + \eta) \overline{g(\tau, \eta)} d\tau d\eta, \quad (60)$$

based on the above correlation operator (60), we can obtain the correlation theorem of the RBiFT:

$$F_B^R \{f \tilde{\star} g\}(\omega, v) = F_B^R(f)(\omega, v) F_B^R \{ \overline{\text{Simp}(g)} \}(-\omega, -v) + F_B^R(f)(-\omega, -v) F_B^R \{ \overline{\text{Perp}(g)} \}(-\omega, -v). \quad (61)$$

6 | APPLICATIONS

6.1 | Biquaternion linear time-invariant system analysis

The QFT for quaternion linear time-invariant (QLTI) system [28, 30] have been studied. It is important for quaternion filter design and color image processing. However, there is not research on how to use the BiQLCT for biquaternion linear time-invariant (BiQLTI) system analysis. In this section, based on the relations between the convolution and the RBiQLCT, the usage of the BiQLCT in the analysis of BiQLTI system is explored.

The BiQLTI in two variables can be represented in terms of a convolution operator, which relates the output of the system to its input as

$$h(x, y) = \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(\tau, \eta) \check{g}(x - \tau, y - \eta) d\tau d\eta e^{-\mu \left(\frac{a_1}{2b_1} x^2 + \frac{a_2}{2b_2} y^2 \right)}, \quad (62)$$

where $f(x, y)$ is the input signal, $h(x, y)$ is the output signal, and $\check{g}(x, y) = g(x, y) e^{\mu \left(\frac{a_1}{2b_1} x^2 + \frac{a_2}{2b_2} y^2 \right)}$, $g(x, y)$ is the impulse response. The pictorial representation for this system is given in Figure 1. Then, from the relations between the RBiQLCT and the convolution operator shown in (35), we can analyze the BiQLTI systems easily in the frequency domain.

Next, we will discuss how to use the RBiQLCT to analyze the combination of BiQLTI systems. In Figures 2 and 3, we show two basic ways to combine the BiQLTI systems.

Case 1: When the BiQLTI systems are combined in parallel, as in Figure 2, the relation between the input and the output can be expressed as

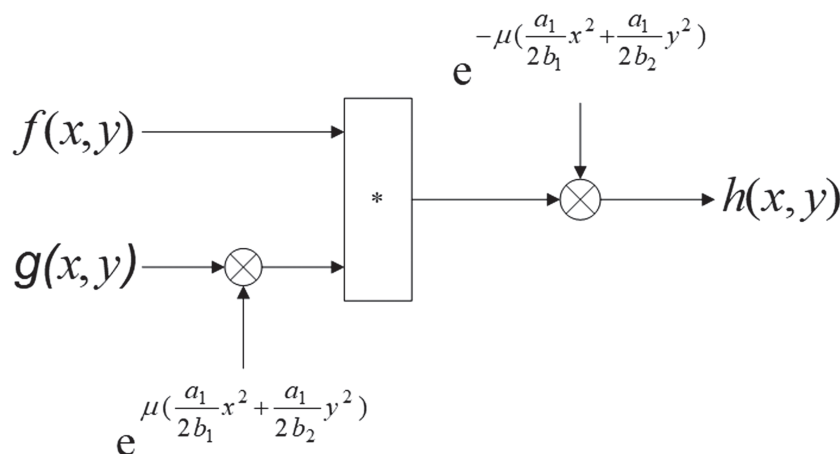


FIGURE 1 Time domain building block for BiQLTI systems.

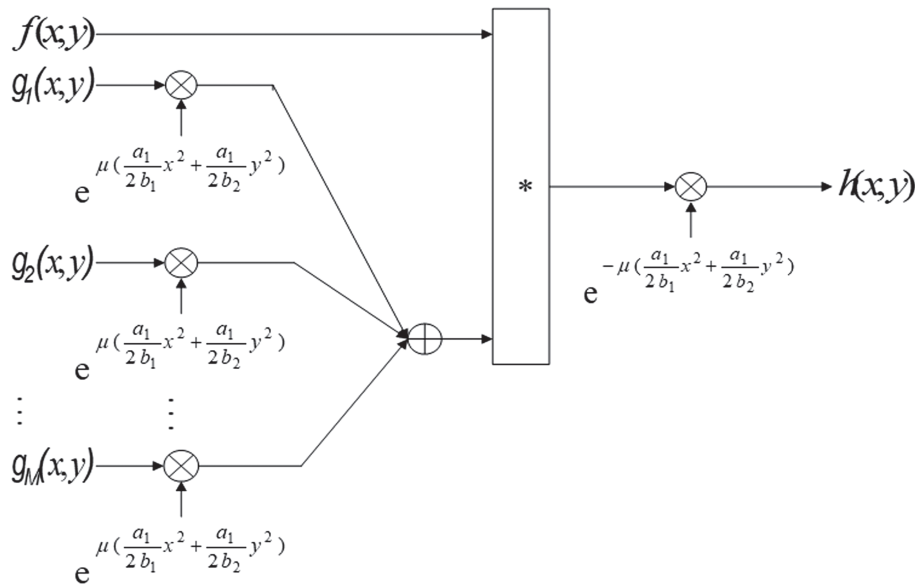


FIGURE 2 Combination of BiQLTI system: In parallel.

$$h(x, y) = [f(x, y) * \check{g}_n(x, y)] e^{-\mu\left(\frac{a_1}{2b_1}x^2 + \frac{a_2}{2b_2}y^2\right)}, \quad (63)$$

where $\check{g}_n(x, y) = \sum_{m=1}^M \check{g}_m(x, y)$ and $\check{g}_m(x, y) = g_m(x, y) e^{\mu\left(\frac{a_1}{2b_1}x^2 + \frac{a_2}{2b_2}y^2\right)}$. In the frequency domain, based on (35), we obtain the following relation:

$$\mathcal{L}_{A_1, A_2}^{RB}\{h\}(\omega, v) = \left[F_B^R(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) F_B^R(f) \left(-\frac{\omega}{b_1}, -\frac{v}{b_2} \right) \right] \left\{ \sum_m^M \left[\begin{matrix} \mathcal{L}_{A_1, A_2}^{RB, m}\{\text{Simp}(g)\}(\omega, v) \\ \mathcal{L}_{A_1, A_2}^{RB, m}\{\text{Perp}(g)\}(\omega, v) \end{matrix} \right] \right\}. \quad (64)$$

Corollary 10. If $A_1 = A_2 = (0, 1, -1, 0)$, then we can obtain the BiQLTI systems in parallel by the RBiQFT, as follows:

$$F_B^R\{h\}(\omega, v) = [F_B^R(f)(\omega, v) F_B^R(f)(-\omega, -v)] \left\{ \sum_m^M \left[\begin{matrix} F_B^{R, m}\{\text{Simp}(g)\}(\omega, v) \\ F_B^{R, m}\{\text{Perp}(g)\}(\omega, v) \end{matrix} \right] \right\}. \quad (65)$$

Case 2: When the BiQLTI systems are combined in series, as in Figure 3, the relation between the input and the output can be expressed as

$$h(x, y) = \left(\left[\left(\dots \left[\left((f(x, y) * \check{g}_1(x, y)) e^{-\mu\left(\frac{a_1}{2b_1}x^2 + \frac{a_2}{2b_2}y^2\right)} \right) * \check{g}_2(x, y) \right] e^{-\mu\left(\frac{a_1}{2b_1}x^2 + \frac{a_2}{2b_2}y^2\right)} \right] \right. \right. \right. \\ \left. \left. \left. * \dots * \check{g}_{M-1}(x, y) \right] e^{-\mu\left(\frac{a_1}{2b_1}x^2 + \frac{a_2}{2b_2}y^2\right)} \right] * \check{g}_M(x, y) \right] e^{-\mu\left(\frac{a_1}{2b_1}x^2 + \frac{a_2}{2b_2}y^2\right)}, \quad (66)$$

where $\check{g}_t(x, y) = g_t(x, y) e^{\mu\left(\frac{a_1}{2b_1}x^2 + \frac{a_2}{2b_2}y^2\right)}$, $t = 1, 2, \dots, M$.

Then, in the frequency domain, we obtain the following relation:

$$\mathcal{L}_{A_1, A_2}^{RB}\{h\}(\omega, v) \\ = \left[F_{e, B}^R(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) F_{o, B}^R(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \right] \left[\begin{matrix} F_{e, B}^{R, 1}\{g\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) F_{o, B}^{R, 1}\{g\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \\ F_{o, B}^{R, 1}\{g'\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) F_{e, B}^{R, 1}\{g'\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \end{matrix} \right] \left[\begin{matrix} F_{e, B}^{R, 2}\{g\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) F_{o, B}^{R, 2}\{g\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \\ F_{o, B}^{R, 2}\{g'\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) F_{e, B}^{R, 2}\{g'\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \end{matrix} \right] \\ \dots \left[\begin{matrix} F_{e, B}^{R, M-1}\{g\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) F_{o, B}^{R, M-1}\{g\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \\ F_{o, B}^{R, M-1}\{g'\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) F_{e, B}^{R, M-1}\{g'\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \end{matrix} \right] \left[\begin{matrix} \mathcal{L}_{A_1, A_2}^{RB, M}\{g\}(\omega, v) \\ \mathcal{L}_{A_1, A_2}^{RB, M}\{g'\}(\omega, v) \end{matrix} \right]. \quad (67)$$

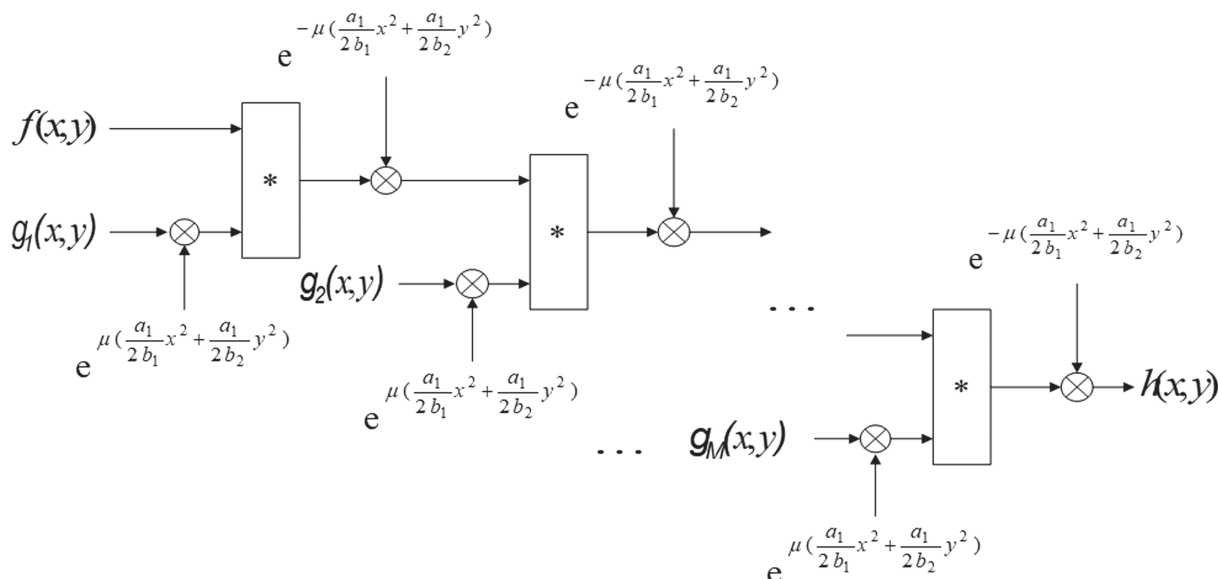


FIGURE 3 Combination of BiQLTI system: In series.

Proof. Let $h_k(x, y)$ ($2 \leq k < M$) be the cascade output of $g_k(x, y)$, $h_{M-1}(x, y)$ be the cascade input of $g_M(x, y)$, and $h(x, y) = h_M(x, y)$ be the output of $g_M(x, y)$, according to (44), then we obtain

$$\begin{aligned} \mathcal{L}_{A_1, A_2}^{RB}\{h\}(\omega, v) &= F_{e, B}^{R, M-1}(h) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \mathcal{L}_{A_1, A_2}^{RB, M}\{g\}(\omega, v) + F_{o, B}^{R, M-1}(h) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \mathcal{L}_{A_1, A_2}^{RB, M}\{g'\}(\omega, v) \\ &= \left[F_{e, B}^{R, M-1}(h) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \quad F_{o, B}^{R, M-1}(h) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \right] \begin{bmatrix} \mathcal{L}_{A_1, A_2}^{RB, M}\{g\}(\omega, v) \\ \mathcal{L}_{A_1, A_2}^{RB, M}\{g'\}(\omega, v) \end{bmatrix}. \end{aligned} \quad (68)$$

On the other hand, $h_{k-1}(x, y)$ is the input of $g_k(x, y)$, and $h_k(x, y)$ is the output of $g_k(x, y)$, from (53) and (54), we obtain

$$\begin{aligned} F_{e, B}^{R, k}\{h\}(\omega, v) &= F_{e, B}^{R, k-1}(f)(\omega, v) F_{e, B}^{R, k}\{g\}(\omega, v) + F_{o, B}^{R, k-1}(f)(\omega, v) F_{o, B}^{R, k}\{g'\}(\omega, v) \\ &= \left[F_{e, B}^{R, k-1}(f)(\omega, v) \quad F_{o, B}^{R, k-1}(f)(\omega, v) \right] \begin{bmatrix} F_{e, B}^{R, k}\{g\}(\omega, v) \\ F_{o, B}^{R, k}\{g'\}(\omega, v) \end{bmatrix}, \end{aligned} \quad (69)$$

$$\begin{aligned} F_{o, B}^{R, k}\{h\}(\omega, v) &= F_{e, B}^{R, k-1}(f)(\omega, v) F_{o, B}^{R, k}\{g\}(\omega, v) + F_{o, B}^{R, k-1}(f)(\omega, v) F_{e, B}^{R, k}\{g'\}(\omega, v) \\ &= \left[F_{e, B}^{R, k-1}(f)(\omega, v) \quad F_{o, B}^{R, k-1}(f)(\omega, v) \right] \begin{bmatrix} F_{o, B}^{R, k}\{g\}(\omega, v) \\ F_{e, B}^{R, k}\{g'\}(\omega, v) \end{bmatrix}. \end{aligned} \quad (70)$$

Hence, for $2 \leq k < M$, we have

$$\begin{aligned} &\left[F_{e, B}^{R, k}\{h\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \quad F_{o, B}^{R, k}\{h\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \right] \\ &= \left[F_{e, B}^{R, k-1}(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \quad F_{o, B}^{R, k-1}(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \right] \begin{bmatrix} F_{e, B}^{R, k}\{g\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) & F_{o, B}^{R, k}\{g\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \\ F_{o, B}^{R, k}\{g'\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) & F_{e, B}^{R, k}\{g'\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \end{bmatrix}, \end{aligned} \quad (71)$$

when $k = 1$, then

$$\begin{aligned} &\left[F_{e, B}^{R, 1}\{h\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \quad F_{o, B}^{R, 1}\{h\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \right] \\ &= \left[F_{e, B}^R(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \quad F_{o, B}^R(f) \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \right] \begin{bmatrix} F_{e, B}^{R, 1}\{g\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) & F_{o, B}^{R, 1}\{g\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \\ F_{o, B}^{R, 1}\{g'\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) & F_{e, B}^{R, 1}\{g'\} \left(\frac{\omega}{b_1}, \frac{v}{b_2} \right) \end{bmatrix}. \end{aligned} \quad (72)$$

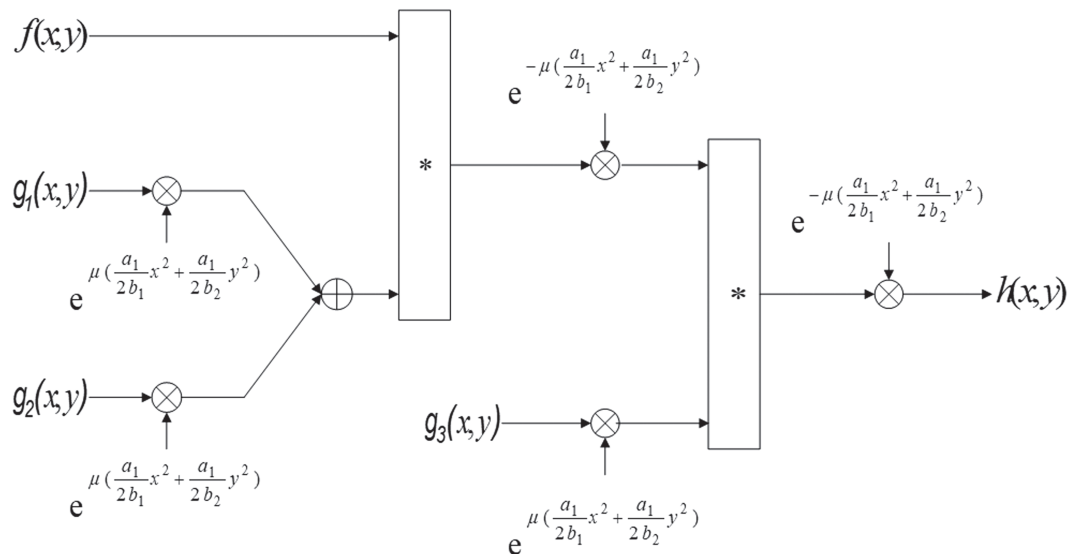


FIGURE 4 Combination of BiQLTI system: In parallel and series.

Substituting (71) into (68) iteratively and together with (72), we can obtain (67). $\square$

Corollary 11. If $A_1 = A_2 = (0, 1, -1, 0)$, then we can obtain the BiQLTI systems in series by the RBiQFT, as follows:

$$F_B^R\{h\}(\omega, v) = \begin{bmatrix} F_{e,B}^R(f)(\omega, v) & F_{o,B}^R(f)(\omega, v) \\ F_{e,B}^{R,1}\{g\}(\omega, v) & F_{e,B}^{R,1}\{g'\}(\omega, v) \\ F_{e,B}^{R,2}\{g\}(\omega, v) & F_{e,B}^{R,2}\{g'\}(\omega, v) \end{bmatrix} \begin{bmatrix} F_{e,B}^{R,M-1}\{g\}(\omega, v) & F_{o,B}^{R,M-1}\{g\}(\omega, v) \\ F_{e,B}^{R,M-1}\{g'\}(\omega, v) & F_{e,B}^{R,M-1}\{g'\}(\omega, v) \end{bmatrix} \begin{bmatrix} F_B^{R,M}\{g\}(\omega, v) \\ F_B^{R,M}\{g'\}(\omega, v) \end{bmatrix}. \quad (73)$$

Applying (64) and (67), we can use the RBiQFT and RBiQLCT to represent many different combinations of BiQLTI systems.

For example, for the BiQLTI systems combined as Figure 4, if we use the RBiQFT and RBiQLCT to represent the above system, then from (64) and (67), we can obtain

$$\mathcal{L}_{A_1, A_2}^{RB}\{h\}(\omega, v) = \begin{bmatrix} F_{e,B}^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) & F_{o,B}^R(f)\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) \end{bmatrix} \begin{bmatrix} F_{e,B}^{R,4}\{g\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) & F_{o,B}^{R,4}\{g\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) \\ F_{e,B}^{R,4}\{g'\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) & F_{e,B}^{R,4}\{g'\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) \end{bmatrix} \begin{bmatrix} \mathcal{L}_{A_1, A_2}^{RB,3}\{g\}(\omega, v) \\ \mathcal{L}_{A_1, A_2}^{RB,3}\{g'\}(\omega, v) \end{bmatrix}, \quad (74)$$

where

$$F_{e,B}^{R,4}\{g\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) = F_{e,B}^{R,1}\{g\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) + F_{e,B}^{R,2}\{g\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right), \quad (75)$$

$$F_{o,B}^{R,4}\{g\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) = F_{o,B}^{R,1}\{g\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) + F_{o,B}^{R,2}\{g\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right), \quad (76)$$

$$F_{e,B}^{R,4}\{g'\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) = F_{e,B}^{R,1}\{g'\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) + F_{e,B}^{R,2}\{g'\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right), \quad (77)$$

$$F_{o,B}^{R,4}\{g'\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) = F_{o,B}^{R,1}\{g'\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right) + F_{o,B}^{R,2}\{g'\}\left(\frac{\omega}{b_1}, \frac{v}{b_2}\right). \quad (78)$$

7 | HEISENBERG UNCERTAINTY PRINCIPLE FOR THE BIQLCT

The Heisenberg uncertainty principles of the QLCT have been studied in [8, 11, 12]. Based on the Heisenberg uncertainty principles of the QLCT, we can obtain the Heisenberg uncertainty principle of the RBiQLCT.

Lemma 1. Let $A_i = \begin{bmatrix} a_i & b_i \\ c_i & d_i \end{bmatrix} \in \mathbb{R}^{2 \times 2}$ be a matrix parameter satisfying $\det(A_i) = 1$, for $i = 1, 2$. If $f \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$, $\omega \mathcal{L}_{A_1, A_2}^{RB}\{f\} \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$, and $v \mathcal{L}_{A_1, A_2}^{RB}\{f\} \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$, then we obtain

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 \left| \mathcal{L}_{A_1, A_2}^{RB}\{f\}(\omega, v) \right|^2 d\omega dv = b_1^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left| \frac{\partial}{\partial x_2} f(x_2, y_2) \right|^2 dx_2 dy_2. \quad (79)$$

and

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} v^2 \left| \mathcal{L}_{A_1, A_2}^{RB}\{f\}(\omega, v) \right|^2 d\omega dv = b_2^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left| \frac{\partial}{\partial x_2} f(x_2, y_2) \right|^2 dx_2 dy_2. \quad (80)$$

Proof. According to the definition of the RBiQLCT, we have

$$\begin{aligned} & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 \left| \mathcal{L}_{A_1, A_2}^{RB}\{f\}(\omega, v) \right|^2 d\omega dv \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 \left(\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, y_1) K_{A_1}^{\mu}(x_1, \omega) K_{A_2}^{\mu}(y_1, v) dx_1 dy_1 \right) \overline{\left(\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_2, y_2) K_{A_1}^{\mu}(x_2, \omega) K_{A_2}^{\mu}(y_2, v) dx_2 dy_2 \right)} d\omega dv, \end{aligned} \quad (81)$$

Based on the biquaternion conjugation, the above formula becomes that

$$\begin{aligned} & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 \left| \mathcal{L}_{A_1, A_2}^{RB}\{f\}(\omega, v) \right|^2 d\omega dv \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 f(x_1, y_1) K_{A_1}^{\mu}(x_1, \omega) \left(K_{A_2}^{\mu}(y_1, v) \overline{K_{A_2}^{\mu}(y_2, v)} \right) \overline{K_{A_1}^{\mu}(x_2, \omega)} \cdot \overline{f(x_2, y_2)} dx_1 dy_1 dx_2 dy_2 d\omega dv \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 f(x_1, y_1) K_{A_1}^{\mu}(x_1, \omega) \left(e^{\mu \frac{a_2}{2b_2} (y_1^2 - y_2^2)} \delta(y_2 - y_1) \right) \overline{K_{A_1}^{\mu}(x_2, \omega)} \cdot \overline{f(x_2, y_2)} dx_1 dy_1 dx_2 dy_2 d\omega \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 \left(\int_{-\infty}^{\infty} f(x_1, y_1) e^{\mu \frac{a_2}{2b_2} (y_1^2 - y_2^2)} \delta(y_2 - y_1) dy_1 \right) K_{A_1}^{\mu}(x_1, \omega) \overline{K_{A_1}^{\mu}(x_2, \omega)} \cdot \overline{f(x_2, y_2)} dx_1 dx_2 dy_2 d\omega \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 f(x_1, y_2) K_{A_1}^{\mu}(x_1, \omega) \overline{K_{A_1}^{\mu}(x_2, \omega)} \cdot \overline{f(x_2, y_2)} dx_1 dx_2 dy_2 d\omega. \end{aligned} \quad (82)$$

On the other hand, applying the definition of the kernel function, we have

$$\begin{aligned} \int_{-\infty}^{\infty} \omega^2 K_{A_1}^{\mu}(x_1, \omega) \overline{K_{A_1}^{\mu}(x_2, \omega)} d\omega &= \frac{1}{2\pi|b_1|} \int_{-\infty}^{\infty} \omega^2 e^{\mu \frac{a_1}{2b_1} (x_1^2 - x_2^2)} \frac{b_1^2}{\omega^2} (-\mu)^{-2} \frac{\partial^2}{\partial x_1^2} e^{-\mu \frac{\omega}{b_1} (x_1 - x_2)} d\omega \\ &= e^{\mu \frac{a_1}{2b_1} (x_1^2 - x_2^2)} b_1^2 \frac{\partial^2}{\partial x_1^2} \frac{1}{2\pi|b_1|} \int_{-\infty}^{\infty} e^{-\mu \frac{\omega}{b_1} (x_1 - x_2)} d\omega \\ &= b_1^2 e^{\mu \frac{a_1}{2b_1} (x_1^2 - x_2^2)} \frac{\partial^2}{\partial x_1^2} \delta(x_1 - x_2). \end{aligned} \quad (83)$$

Hence, (82) reduced that

$$\begin{aligned}
 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 \left| \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) \right|^2 d\omega dv &= b_1^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_1, y_2) e^{\mu_{\frac{a_1}{2b_1}}(x_1^2 - x_2^2)} \frac{\partial^2}{\partial x_1^2} \delta(x_1 - x_2) \overline{f(x_2, y_2)} dx_1 dx_2 dy_2 \\
 &= b_1^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} f(x_2, y_2) \frac{\partial^2}{\partial x_2^2} \overline{f(x_2, y_2)} dx_2 dy_2 \\
 &= b_1^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left| \frac{\partial}{\partial x_2} f(x_2, y_2) \right|^2 dx_2 dy_2.
 \end{aligned} \tag{84}$$

Applying the same method, we can obtain

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} v^2 \left| \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) \right|^2 d\omega dv = b_2^2 \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left| \frac{\partial}{\partial x_2} f(x_2, y_2) \right|^2 dx_2 dy_2. \tag{85}$$

□

Theorem 12. Assume that $f \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$, $\omega \mathcal{L}_{A_1, A_2}^{RB} \{f\} \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$, and $v \mathcal{L}_{A_1, A_2}^{RB} \{f\} \in L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})$, then the following inequalities hold:

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 |f(x, y)|^2 dx dy \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 \left| \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) \right|^2 d\omega dv \geq \frac{b_1^2}{4} \|f\|_{L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})}^4, \tag{86}$$

and

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y^2 |f(x, y)|^2 dx dy \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} v^2 \left| \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) \right|^2 d\omega dv \geq \frac{b_2^2}{4} \|f\|_{L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})}^4. \tag{87}$$

Proof. According to Lemma 1 and the Cauchy–Schwarz inequality, we obtain

$$\begin{aligned}
 &\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 |f(x, y)|^2 dx dy \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 \left| \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) \right|^2 d\omega dv \\
 &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 |f(x, y)|^2 dx dy \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left| b_1 \frac{\partial}{\partial x} f(x, y) \right|^2 dx dy \\
 &\geq \frac{b_1^2}{4} \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(\frac{\partial}{\partial x} f(x, y) \overline{x f(x, y)} + x f(x, y) \overline{\frac{\partial}{\partial x} f(x, y)} \right) dx dy \right]^2 \\
 &= \frac{b_1^2}{4} \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x \frac{\partial}{\partial x} \left(f(x, y) \overline{f(x, y)} \right) dx dy \right]^2 \\
 &= \frac{b_1^2}{4} \left[\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \left(x \frac{\partial}{\partial x} |f(x, y)|^2 \right) dx dy \right]^2.
 \end{aligned} \tag{88}$$

Using integration by parts, the above formula becomes

$$\begin{aligned}
 & \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x^2 |f(x, y)|^2 dx dy \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} \omega^2 \left| \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) \right|^2 d\omega dv \\
 & \geq \frac{b_1^2}{4} \left[- \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} |f(x, y)|^2 dx dy \right]^2 \\
 & = \frac{b_1^2}{4} \|f\|_{L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})}^4.
 \end{aligned} \tag{89}$$

We can take similar steps as above and obtain

$$\int_{-\infty}^{\infty} \int_{-\infty}^{\infty} y^2 |f(x, y)|^2 dx dy \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} v^2 \left| \mathcal{L}_{A_1, A_2}^{RB} \{f\}(\omega, v) \right|^2 d\omega dv \geq \frac{b_2^2}{4} \|f\|_{L^2(\mathbb{R}^2, \mathbb{H}_{\mathbb{C}})}^4.$$

□

8 | CONCLUSIONS

In this paper, we proposed a new transform tool—the BiQLCT. Three definitions are given: LBiQLCT, TBiQLCT, and RBiQLCT. Then, the relationships between them are obtained. Based on the RBiQLCT, some general properties of this transform are proved. In addition, the convolution and correlation theorems associated with the RBiQLCT are studied. According to the convolution operator and convolution theorem, the biquaternion linear time-invariant system is analyzed. Finally, the Heisenberg uncertainty principle for the RBiQLCT is established.

AUTHOR CONTRIBUTIONS

Wen-Biao Gao: Conceptualization; methodology; investigation; validation; writing—review & editing; writing—original draft. **Bing-Zhao Li:** Funding acquisition.

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CONFLICT OF INTEREST STATEMENT

This work does not have any conflicts of interest.

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